Right Triangles Maze ANSWER KEY

Start top-left. Exact answers in simplest radical form; round trig answers to the nearest tenth. Go through the door with your answer.

START

1 Find the hypotenuse.
leg 6, leg 8

48

Find the hypotenuse.
leg 9, leg 12

$3\sqrt{7}$

30°-60°-90° triangle. Find the hypotenuse.
long leg $7\sqrt{3}$

$\sqrt{674}$

Find the other leg.
hypotenuse 25
leg 7

10

$2\sqrt{7}$

21

7

12.9

$14\sqrt{3}$

18

2 Find the other leg.
hypotenuse 13
leg 5

12

3 Find the hypotenuse.
leg 4, leg 6

2.5

Find the hypotenuse.
angle 25°
opposite leg 6

6.6

45°-45°-90° triangle. Find the hypotenuse.
leg $3\sqrt{2}$

$\sqrt{194}$

$2\sqrt{13}$

$4\sqrt{13}$

$2\sqrt{5}$

7.7

6.4

30.3°

4 Find the hypotenuse.
45°-45°-90° triangle
leg 7

14

6 Find the long leg.
30°-60°-90° triangle
short leg 4

8

9 Find the opposite leg.
angle 40°
adjacent leg 10

8.4

10 Find the angle opposite the leg of 7.
opposite leg 7
hypotenuse 12

$7\sqrt{2}$

$5\sqrt{2}$

$4\sqrt{3}$

11.9

6.2

54.3°

35.7°

5 Find a leg.
45°-45°-90° triangle
hypotenuse 10

$10\sqrt{2}$

7 Find the long leg.
30°-60°-90° triangle
hypotenuse 18

$9\sqrt{3}$

8 Find the opposite leg.
angle 31°
hypotenuse 12

10.3

FINISH

You made it!

Right Triangles Maze: Answer Grid **ANSWER KEY**

Write each answer in the order you found it, and show your work for that room.

1	Answer: **10**	$6^2 + 8^2 = c^2 \rightarrow 100 = c^2 \rightarrow c = 10$
2	Answer: **12**	$5^2 + b^2 = 13^2 \rightarrow b^2 = 144 \rightarrow b = 12$
3	Answer: **$2\sqrt{13}$**	$4^2 + 6^2 = c^2 \rightarrow c^2 = 52 \rightarrow c = 2\sqrt{13}$
4	Answer: **$7\sqrt{2}$**	$\text{hyp} = \text{leg} \cdot \sqrt{2} \rightarrow \text{hyp} = 7\sqrt{2}$
5	Answer: **$5\sqrt{2}$**	$\text{leg} = \frac{10}{\sqrt{2}} \rightarrow \text{leg} = 5\sqrt{2}$
6	Answer: **$4\sqrt{3}$**	$\text{long} = \text{short} \cdot \sqrt{3} \rightarrow \text{long} = 4\sqrt{3}$
7	Answer: **$9\sqrt{3}$**	$\text{short} = 9 \rightarrow \text{long} = 9\sqrt{3}$
8	Answer: **6.2**	$\sin 31° = \frac{x}{12} \rightarrow x = 12 \sin 31° \rightarrow x \approx 6.2$
9	Answer: **8.4**	$\tan 40° = \frac{x}{10} \rightarrow x = 10 \tan 40° \rightarrow x \approx 8.4$
10	Answer: **35.7°**	$\sin \theta = \frac{7}{12} \rightarrow \theta = \sin^{-1}\left(\frac{7}{12}\right) \rightarrow \theta \approx 35.7°$